

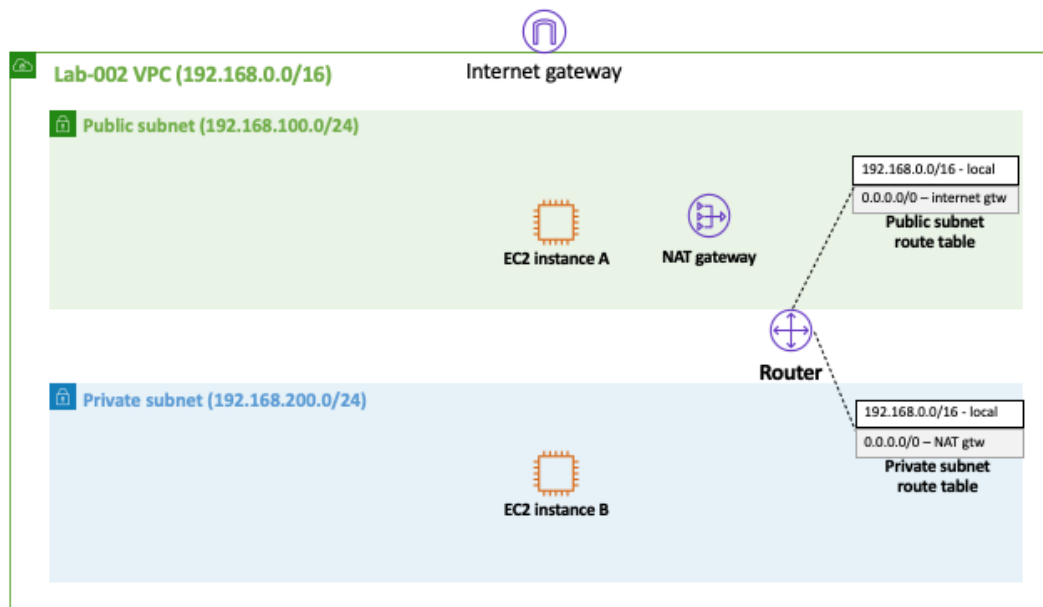
Lab-003

A Single EC2 Instance in a Private Subnet + Bastion Host + NAT Gateway

Goal

This lab illustrates how an EC2 instance running in a *private* subnet can be configured to access the internet with the help of a service called [NAT gateway](#).

Architecture Diagram



Overview

In order to achieve the goal of this lab, you will have to go through the following steps:

Step 1 - Run Lab-002

Repeat all of the steps described in [lab-002](#), making sure that any reference to lab-002 are replaced by lab-003.

Step 2 - Create a NAT Gateway

Create a [NAT gateway](#) in the public subnet. Note that an elastic public IP must be allocated before a NAT gateway can be created

Create NAT Gateway

Actions ▾

Filter by tags and attributes or search by keyword

⏮

<

None found

>

⏭

You do not have any NAT Gateways in this region

Click the Create NAT Gateway button to create your first NAT Gateway

Create NAT Gateway

[NAT Gateways](#) > Create NAT Gateway

Create NAT Gateway

Create a NAT gateway and assign it an Elastic IP address. [Learn more.](#)

Subnet*

Search subnets by ID or name or VPC e.g. "subnet-1a2b3c4d"

⌂ ⓘ

Elastic IP Allocation ID*

Filter by attributes

Subnet ID	Subnet Name	VPC ID	VPC Name
subnet-63b57d05	private	vpc-924e57f5	default
subnet-b1b256eb	public	vpc-924e57f5	default
subnet-0b9f50af8cac77468	public	vpc-0d8181c524284e897	lab-003
subnet-00490a6d5b8f92f1d	private	vpc-0d8181c524284e897	lab-003

Key (128 characters maximum)

Value (256 characters maximum)

Name

lab-003

✕

Add Tag

49 remaining (Up to 50 tags maximum)

* Required

Cancel

Create a NAT Gateway

[NAT Gateways](#) > Create NAT Gateway

Create NAT Gateway

✓

Your NAT gateway has been created.

Note: In order to use your NAT gateway, ensure that you [edit your route tables](#) to include a route with the following NAT gateway. [Find out more.](#)

NAT Gateway ID

nat-025daf83e4e60d9d3

Edit route tables

Close

Step 3 - Create a Route Table

Create a new route table with the *Name tag* private and with a default route to the NAT gateway created in step 2.

Create route table

Actions

Filter by tags and attributes or search by keyword

1 to 3 of 3

☐	Name	Route Table ID	Explicit subnet association	Edge associations
☐	main	rtb-03cb78a03f95efc35	subnet-00490a6d5b8f92f1d	-
☐	public	rtb-0d297ed554058cbb5	subnet-0b9f50af8cac77468	-
☐		rtb-a31287c5	-	-

Route Tables > Create route table

Create route table

A route table specifies how packets are forwarded between the subnets within your VPC, the internet, and your VPN connection.

Name tag

private

VPC*

* Required

Cancel

Create

Route Tables > Create route table

Create route table

✓

The following Route Table was created:

Route Table ID

rtb-0ca0c2010b0440518

Close

Create route table

Actions

Filter by tags and attributes or search by keyword

<

1 to 4 of 4

>

☐	Name	Route Table ID	Explicit subnet association	Edge associations	Main
☐	main	rtb-03cb78a03f95efc35	subnet-00490a6d5b8f92f1d	-	Yes
☒	private	rtb-0ca0c2010b0440518	-	-	No
☐	public	rtb-0d297ed554058cbb5	subnet-0b9f50af8cac77468	-	No
☐		rtb-a31287c5	-	-	Yes

Route Table: rtb-0ca0c2010b0440518

Summary

Routes

Subnet Associations

Edge Associations

Route Propagation

Tags

Edit routes

View

All routes

Destination	Target	Status
192.168.0.0/16	local	active

Create route table

Actions

Filter by tags and attributes or search by keyword

<

1 to 4 of 4

>

☐	Name	Route Table ID	Explicit subnet association	Edge associations	Main
☐	main	rtb-03cb78a03f95efc35	subnet-00490a6d5b8f92f1d	-	Yes
☒	private	rtb-0ca0c2010b0440518	-	-	No
☐	public	rtb-0d297ed554058cbb5	subnet-0b9f50af8cac77468	-	No
☐		rtb-a31287c5	-	-	Yes

Route Table: rtb-0ca0c2010b0440518

Summary

Routes

Subnet Associations

Edge Associations

Route Propagation

Tags

Edit routes

View

All routes

Destination	Target	Status
192.168.0.0/16	local	active

[Route Tables](#) > Edit routes

Edit routes

Destination	Target	Status	Propagated
192.168.0.0/16	local	active	No
0.0.0.0/0			No

Add route

* Required

- Egress Only Internet Gateway
- Instance
- Internet Gateway
- NAT Gateway
- Network Interface
- Outpost Local Gateway
- Peering Connection
- Transit Gateway

Cancel

Save routes

[Route Tables](#) > Edit routes

Edit routes

✓ Routes successfully edited

Close

[Route Tables](#) > Edit routes

Edit routes

Destination	Target	Status	Propagated
192.168.0.0/16	local	active	No
0.0.0.0/0	nat-		No

Add route

* Required

- nat-025daf83e4e60d9d3 lab-003

Cancel

Save routes

[Route Tables](#) > Edit routes

Edit routes

✓ Routes successfully edited

Close

Step 4 - Associate the Route Table to the Private Subnet

Associate the newly created route table to the private subnet.

Create subnet

Actions

Filter by tags and attributes or search by keyword

1 to 4 of 4

☐	Name	Subnet ID	State	VPC	IPv4 CIDR	Avai
☒	private	subnet-00490a6d5b8f92f1d	available	vpc-0d8181c524284e897 ...	192.168.200....	250
☐	public	subnet-0b9f50af8cac77468	available	vpc-0d8181c524284e897 ...	192.168.100....	249
☐	private	subnet-63b57d05	available	vpc-924e57f5 default	172.31.16.0/20	409C
☐	public	subnet-b1b256eb	available	vpc-924e57f5 default	172.31.0.0/20	409C

Subnet: subnet-00490a6d5b8f92f1d

- Description
- Flow Logs
- Route Table
- Network ACL
- Tags
- Sharing

Edit route table association

Route Table: rtb-03cb78a03f95efc35 | main

1 to 1 of 1

Destination	Target
192.168.0.0/16	local

Subnets > Edit route table association

Edit route table association

Subnet ID subnet-00490a6d5b8f92f1d

Route Table ID* rtb-03cb78a03f95efc35

Filter by attributes

Route table ID	Route table name	VPC ID
rtb-03cb78a03f95efc35	main	vpc-0d8181c524284e897
rtb-0ca0c2010b0440518	private	vpc-0d8181c524284e897
rtb-0d297ed554058cbb5	public	vpc-0d8181c524284e897

Edit route table association

Subnet ID subnet-00490a6d5b8f92f1d

Route Table ID* 

⏪ < 1 to 2 of 2 > ⏩

Destination	Target
192.168.0.0/16	local
0.0.0.0/0	nat-025daf83e4e60d9d3

* Required

Cancel

Save

Edit route table association



Successfully edited route table association

Close

Test and Validation

To validate this lab you need to access the EC2 instance in the private subnet (labeled as B) and from there try to access the internet, for example:

```
curl www.google.com
```